

Supraja Mamidi

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SUMMARY

Data Analyst with 3+ years of experience specializing in healthcare, pharmaceutical, and clinical data analytics. Skilled in designing ETL pipelines, building predictive models, and developing dashboards that deliver actionable insights. Proficient in Python, SQL, R, and cloud-based data platforms, with expertise in healthcare standards including FHIR and HL7. Strong background in quality systems, compliance, and process control, ensuring data integrity and regulatory alignment. Adept at combining analytics with quality improvement to drive measurable business impact.

TECHNICAL SKILLS

Programming & Scripting: Python, SQL, R, SAS, Bash, PowerShell, PySpark

Databases & Data Warehousing: MySQL, SQL Server, PostgreSQL, MongoDB, Cassandra, Snowflake

Data Engineering & ETL: Apache Airflow, SSIS, AWS Glue, Azure Data Factory, DBT

Cloud & Big Data Technologies: AWS (EC2, S3, Lambda, RDS, Redshift, Glue, QuickSight, CloudWatch)

Data Visualization & Business Intelligence: Tableau, Power BI, Looker, SSRS, Microsoft Excel – Pivot Tables, VLOOKUP, Formulas, Plotly, QlikView, Google Data Studio, Matplotlib, Seaborn

Analytics & Machine Learning: Data Mining, Statistical Analysis, Predictive Modeling, A/B Testing, Text Mining, Regression Models, Classification, Clustering, Time Series Forecasting, Natural Language Processing (NLP), Deep Learning

Healthcare Data Standards: Fast Healthcare Interoperability Resources (FHIR), Health Level Seven (HL7)

Quality Systems & Compliance: GxP, ISO 9001, FDA CFR 21 Part 11, CAPA, Root Cause Analysis, Six Sigma (DMAIC), Statistical Process Control (SPC)

Soft Skills: Critical Thinking, Problem-Solving, Communication, Business Acumen, Leadership

PROFESSIONAL EXPERIENCE

Data Analyst, AbbVie

Aug 2024 – Present | Remote, USA

- Designed Snowflake pipelines processing 5TB+ pharmacovigilance data, improving cross-study reporting accuracy by 31% and accelerating patient safety monitoring across multiple oncology and immunology therapeutic portfolios.
- Built Tableau dashboards visualizing 60+ regulatory KPIs, improving compliance oversight by 29% while enabling real-time monitoring of patient dropout, enrollment rates, and treatment adherence across global trial ecosystems.
- Developed machine learning models in Python analyzing 1.8M records, achieving 85% accuracy forecasting adverse outcomes, enabling proactive interventions and supporting data-driven clinical development pipeline decisions.
- Implemented HIPAA-compliant security frameworks across 12 therapeutic datasets, reducing regulatory audit exceptions by 36% while ensuring continuous adherence to FDA CFR 21 Part 11 governance requirements.
- Automated ETL workflows using Airflow and AWS Glue, streamlining 140+ recurring jobs, reducing manual intervention by 41%, and ensuring reliable clinical data delivery across AbbVie's research environments.
- Deployed NLP pipelines with SpaCy analyzing 450K physician notes, extracting safety signals with 23% improved accuracy, strengthening pharmacovigilance reporting and enhancing enterprise-wide patient safety compliance.

Data Analyst, Dr. Reddy's Laboratories

Jul 2022 – Jul 2023 | Remote, India

- Designed Python and SQL pipelines processing 3.1M+ laboratory records, reducing clinical reporting timelines by 37% and improving data accessibility for global pharmaceutical regulatory submissions.
- Built Power BI dashboards tracking 45+ product quality KPIs, improving defect detection speed by 32% and enabling early insights into manufacturing compliance performance across multiple global plants.
- Applied time-series forecasting models in R on 2.6M batch datasets, predicting production variability with 84% accuracy, optimizing resource planning and reducing wastage in pharmaceutical operations.
- Developed automated data integration using Azure Data Factory consolidating 12TB+ clinical, manufacturing, and laboratory records, reducing manual reporting workloads by 34% and strengthening compliance-driven data accessibility.
- Implemented Six Sigma DMAIC methodology on recurring deviations, identifying 21% reduction opportunities in pharmaceutical testing errors and improving production cycle reliability across laboratory environments.
- Established SPC frameworks in SAS and Excel monitoring 65+ variables, minimizing batch instability by 24% while improving regulatory compliance during long-term pharmaceutical stability studies.
- Automated CAPA dashboards in Tableau linking 300+ audit trail records with remediation timelines, achieving 36% faster incident resolution and improving compliance across multiple manufacturing and quality divisions.
- Integrated FHIR and HL7 standards into ETL workflows across 15 systems, enhancing interoperability, improving patient safety reporting reliability by 28%, and ensuring accuracy in regulatory compliance submissions.

EDUCATION

Master of Science, Governors State University

Aug 2023 - May 2025 | Chicago, USA

Health Informatics

Bachelor of Pharmacy, ST. Mary's College of Pharmacy affiliated to JNTUH

Aug 2018 – Jul 2022 | Hyderabad, India

Pharmacy